Experiments for Smooth RS-MCA Proximity Thresholds

Fresh-Eyes Threshold Memo, Missing Theorems, and Retained Locator Ledger

RS-MCA experimental notes

June 26, 2026

Abstract

This replacement starts with the current "fresh eyes" synthesis for the smooth Reed–Solomon Proximity Prize problem. The negative side is now quite rigid: Paper D gives a prize-facing upper cap on the MCA threshold, conditional on the Crites–Stewart list-to-agreement import, while Paper A and the list lower bounds rule out any no-slack capacity theorem. The positive side is not yet a theorem; it should be organized around two local-limit problems, L1 arbitrary-word locator fibers and M1 all-line residue packing, plus extension-line MCA, the Crites–Stewart import audit, and exact protocol-ledger rewrites.

The main new additions over the old `experiments.tex` are two compact bridge lemmas. First, for ordinary column-distance interleaving, interleaved lists inject into common feasible supports, so a uniform base locator-fiber theorem gives the same polynomial exponent for all protocol arities; there is no automatic $n^{\mu B}$ product loss. Second, a list bound plus an MCA bound implies a concrete line-decoding bound with loss $M + nL + 1$. The older BCHKS provenance notes, quotient-locator dictionaries, normal forms, arithmetic gates, and finite-threshold infrastructure are retained below whenever they still look useful.

Contents

1 Fresh-Eyes Settlement Memo **2**
1.1 Bottom line 3
1.2 What is already proved or very close to proved 4
1.3 The corrected positive theorem shape 4
1.4 What remains missing 4

2 New Bridge I: Common-Support Interleaved Fiber Injection **5**

3 New Bridge II: List plus MCA Gives Line Decoding **6**

4 Updated Positive-Theorem Package **7**

5 Immediate Experimental Additions **7**

6 Fresh Non-Claims **8**

7 Retained Previous Experiments Material **8**

8 Status, Provenance, and What Counts as New **8**

9 BCHKS Locator Construction in Smooth-Quotient Notation **9**
9.1 Imported theorem/proof pattern . . . 9
9.2 Notation dictionary . . . 11
9.3 The shared locator identity . . . 11
9.4 List-fiber pigeonhole wrapper . . . 12
9.5 Slack-two and subfield target for Paper D . . . 12
9.6 Pasteable main-paper wording . . . 12

10 Triage of the AI-Generated Four-Item Packet **13**

11 Restricted Resonance Setup for Paper B **14**

12 Cycle 14 Normal Form **15**

13 Cycle 18 Component Split **16**

14 New Divisibility-Gate Theorem **16**

15 Nearby Proven Gates **17**

16 Slope Map and Remaining Heuristics **18**

17 Next Experimental Checks **18**

18 June 26 Normal-Form and Dependency Results **18**
18.1 Syndrome-pencil normal form for support-wise line incidence . . . 19
18.2 Interleaved codegree reduction . . . 20
18.3 Deep-point cap dependency split . . . 20
18.4 Common code-line residual budget . . . 21

19 High-Agreement Tangent Staircase **22**

20 High-Agreement Adjacent Ledgers **26**
20.1 No-loss CA and projective slopes . . . 26
20.2 Finite-parameter curve MCA . . . 27
20.3 Interleaved-list uniqueness . . . 28
20.4 The $\mathbb{F}_{17^{32}}$, $n = 512$, $k = 256$ protocol-facing ledger . . . 29

21 Non-Claims **33**

A BibTeX Entry to Copy into Main Papers **34**

1 Fresh-Eyes Settlement Memo

I would frame the current state as follows. The attached suite substantially settles the negative side and gives a plausible corrected theorem shape, but it does not yet settle the Proximity Prize threshold in the sharp "largest δ^*" sense. The clean prize-relevant statement is no longer "prove smooth RS reaches capacity". It is

$$\delta = 1 - \rho - \eta, \qquad \eta \text{ must clear generated-field entropy, quotient-core, MCA-floor,}$$

interleaving, line-field, and challenge-field budgets.

For

$$C = \mathrm{RS}[\mathbb{F}, D, k], \qquad n = |D|, \qquad \rho = k/n \in \{1/2, 1/4, 1/8, 1/16\},$$

the official challenge statements use this MCA/list-threshold form; in this experimental ledger the concrete prize envelope is [1]

$$\varepsilon_* = 2^{-128}, \qquad k \le 2^{40}, \qquad |\mathbb{F}| < 2^{256}.$$

The protocol-relevant soundness contribution has the schematic form

$$\varepsilon_{\mathrm{mca}}(C, \delta) + \frac{|\Lambda(C^{\equiv\mu}, \delta)|}{q_{\mathrm{line}}} + \text{query error.}$$

Thus a list theorem alone is not enough unless it is connected to the exact MCA, correlated-agreement, line-decoding, or curve-MCA quantity used by the protocol, and unless the denominator is the actual line or challenge field.

1.1 Bottom line

For the **MCA/correlated-agreement threshold**, the strongest attached result is Paper D's universal cap. Conditional on the direct Crites–Stewart conversion and the stated dyadic-divisor hypotheses,

$$\delta^*_{\mathrm{MCA}}(2^{-128}) \le 1 - \rho - 2^{-9} \quad (\rho = 1/2, 1/4, 1/8),$$

and

$$\delta^*_{\mathrm{MCA}}(2^{-128}) \le 1 - \rho - 2^{-10} \quad (\rho = 1/16).$$

Equivalently, throughout the relevant unsafe intervals Paper D gives

$$\varepsilon_{\mathrm{mca}}(C, \delta) > \frac{1}{2k}\left(1 - \frac{n}{|\mathbb{F}|}\right) \ge 2^{-86} \gg 2^{-128}.$$

This is a real prize-facing cap, not merely a heuristic obstruction. It is not an error-one theorem; it proves failure probability on the order of $1/k$, which is already far above 2^{-128}, but it does not characterize the full failure curve.

For the **interleaved-list threshold**, Paper A's base list lower bound plus the base-to-interleaved monotonicity gives a clean negative cap. At quotient scale $N \mid n$, with $\rho N \in \mathbb{Z}$, there is a received word with at least

$$\binom{N}{\rho N + 1}/q, \qquad q = |\mathbb{F}|,$$

base-code codewords at radius $1 - \rho - 1/N$. Therefore the same radius is unsafe for the interleaved-list budget whenever this exceeds $2^{-128}q$. In the worst case $q < 2^{256}$, the following choices suffice:

ρ	N	list-failure radius	$\log_2 \binom{N}{\rho N+1} - 256$	margin over 2^{128}
1/2	512	$1/2 - 2^{-9}$	251.17	123.17
1/4	512	$3/4 - 2^{-9}$	156.33	28.33
1/8	1024	$7/8 - 2^{-10}$	298.68	170.68
1/16	2048	$15/16 - 2^{-11}$	433.89	305.89

The fourth column is the exponent of the lower-bound list size after dividing by the worst-case field size 2^{256}; the fifth subtracts the target exponent 128.

1.2 What is already proved or very close to proved

The old no-slack smooth-domain capacity claim is dead. The quotient-locator construction gives explicit bad slopes for canonical lines and, over deployed smooth prime fields such as BabyBear, KoalaBear, and $3\cdot 2^{30}+1$, yields support-wise line-MCA error one on intervals starting at $1-\rho-2^{-18}$ for all four prize rates, with the sharper 2^{-17} gap at rates 1/2 and 1/4. These statements rule out any theorem saying that every radius below $1-\rho$ has negligible MCA error.

Paper D is the most prize-relevant negative theorem because it converts a list phenomenon into CA and then MCA at the exact 2^{-128}-style threshold. Its load-bearing import is the Crites–Stewart list-to-agreement conversion. The import must be audited, but the use is shallow: integer-radius compatibility, the augmented-code list conclusion, normalization of $\varepsilon_{\mathrm{ca}}$, domain/coset coverage, and the CA-to-MCA monotonicity step.

The list lower-bound side is also stronger than it is currently advertised. It should be promoted to an explicit interleaved-list negative corollary: base lists embed into ordinary interleaved lists, and the binomial numerators above already exceed the $2^{-128}|\mathbb{F}|$ budget in the full field envelope for the displayed quotient divisors.

1.3 The corrected positive theorem shape

The positive theorem should use

$$a = \lceil (1-\delta)n \rceil = k+\sigma, \qquad \delta = 1-\rho-\eta,$$

and require at least

$$\sigma \log_2 q_{\mathrm{gen}} \geq (1+\varepsilon)\log_2 \binom{n}{k+\sigma}, \qquad \sigma \geq C\frac{n}{\log n},$$

plus explicit quotient-profile and failure-floor ledgers. The first inequality uses the generated field q_{gen}, not a larger extension or challenge field that merely appears later in the protocol. The MCA numerator must be divided by the actual line field q_{line}.

The intended list target is an arbitrary-word locator-fiber theorem

$$\sup_U |\operatorname{Fib}_U(k+\sigma)| \leq n^B.$$

The intended MCA target is an all-line bound of the form

$$\varepsilon_{\mathrm{mca}}(C, 1-\rho-\eta) \leq \frac{n^{1+o(1)} + 2^{(\beta(\rho)/H_2(\rho))\mathcal{Q}_H(\eta)(1+o(1))}}{q_{\mathrm{line}}},$$

where quotient-periodic classes are removed or charged by $\mathcal{Q}_H(\eta)$. The current packet contains important special cases and reductions, but not these full arbitrary-word and all-line theorems.

1.4 What remains missing

1. **L1: arbitrary-word locator local limit.** This is the real list upper-bound theorem. Characteristic-zero rigidity, monomial-prefix lanes, and split-prime transfer results are useful but do not yet imply the uniform polynomial-field theorem needed for arbitrary received words.

2. **M1: all-line aperiodic residue packing.** Canonical-line exactness and quotient floors are lower-bound tools, not all-line upper bounds. The positive MCA theorem needs a fixed-field, all-line, all-base residue-packing bound.

3. **F1: extension-line MCA.** The list side has a clean extension-code identity. MCA does not. Base-valued bad lines can deflate when sampled over an extension field, while Paper D's cap indicates that genuinely extension-valued bad lines exist in relevant regimes. Any extension-line lift must therefore be a corrected-reserve theorem, not a blanket subfield statement.

4. **Crites–Stewart import audit.** Paper D's universal cap is load-bearing on the direct conversion. The integer-radius condition, augmented-code list conclusion, normalization of $\varepsilon_{\mathrm{ca}}$, and domain/coset hypotheses should be independently checked.

5. **Protocol-ledger rewrite.** Deployed FRI/WHIR/IOP analyses must be rewritten with exact terms
$$|\Lambda(C^{\equiv\mu}, \delta)|/q_{\mathrm{line}} \quad \text{and} \quad \varepsilon_{\mathrm{mca}} \text{ or line-decoding over the actual line field.}$$
The survey toy protocol already has this shape; deployed analyses still need the same ledger.

2 New Bridge I: Common-Support Interleaved Fiber Injection

Status: New-local wrapper. The older ledger treated interleaving conservatively through a safe product bound $L_\mu(\delta) \le L_1(\delta)^\mu$. For ordinary column-distance RS interleaving, the following support-level injection gives the sharper upper bound: once the one-row locator-fiber theorem is proved, interleaving costs no extra exponent.

For $S \subseteq D$, let
$$L_S(X) = \prod_{x\in S}(X - x).$$
For a received word $U \in \mathbb{F}^D$, the condition $\deg(U \bmod L_S) < k$ means that the unique polynomial of degree less than $|S|$ interpolating $U|_S$ has degree less than k.

Definition 1 (one-row and common interleaved fibers)**.** *Let $C = \mathrm{RS}[\mathbb{F}, D, k]$, let $a \ge k$, and let $\mathbf{U} = (U_1, \dots, U_\mu) \in (\mathbb{F}^\mu)^D$. Define*
$$\mathrm{Fib}^{(\mu)}_{\mathbf{U}}(a) = \{S \subseteq D : |S| = a,\ \deg(U_i \bmod L_S) < k \text{ for every } i = 1, \dots, \mu\}.$$
For one row, write $\mathrm{Fib}_{U_i}(a)$ for the same condition with only U_i.

Lemma 1 (common-support interleaved fiber injection)**.** *Let $C = \mathrm{RS}[\mathbb{F}, D, k]$, $n = |D|$, and let $a = \lceil (1-\delta)n\rceil \ge k$. For every received μ-row word $\mathbf{U} \in (\mathbb{F}^\mu)^D$, ordinary column-distance interleaving satisfies*
$$|\Lambda(C^{\equiv\mu}, \delta, \mathbf{U})| \le |\mathrm{Fib}^{(\mu)}_{\mathbf{U}}(a)| \le \min_{1\le i\le\mu} |\mathrm{Fib}_{U_i}(a)|.$$
Consequently,
$$\sup_{\mathbf{U}} |\Lambda(C^{\equiv\mu}, \delta, \mathbf{U})| \le \sup_U |\mathrm{Fib}_U(a)|.$$
In particular, if L1 proves $\sup_U |\mathrm{Fib}_U(k+\sigma)| \le n^B$, then
$$|\Lambda(C^{\equiv\mu}, 1-\rho-\sigma/n)| \le n^B$$
for every ordinary column-distance protocol arity μ.

Proof. Fix a total order on D. A listed interleaved codeword $(P_1, \ldots, P_\mu) \in C^{\equiv\mu}$ at column distance at most δ from $\mathbf{U}$ agrees with $\mathbf{U}$ on at least a common columns. Choose the first a-subset S of those agreeing columns. For this S, every row $U_i|_S$ is interpolated by the degree-$< k$ polynomial P_i, hence $S \in \mathrm{Fib}_{\mathbf{U}}^{(\mu)}(a)$.

For fixed S, the tuple $(P_1, \ldots, P_\mu)$ is uniquely determined, because each P_i has degree less than k and is fixed by its values on $|S| \geq k$ points. Thus the canonical-support map from listed interleaved codewords to common feasible supports is injective. The second inequality is immediate because every common feasible support is feasible for each individual row. Taking suprema gives the final claim. ☐

Remark 1 (impact)**.** *For ordinary column-distance RS interleaving, L2 should be demoted from a separate product-exponent problem to a corollary of L1. The hard list theorem remains the base arbitrary-word locator local limit.*

3 New Bridge II: List plus MCA Gives Line Decoding

Status: New-local bridge. The next proposition turns a line-decoding proof obligation into a finite-constant budget once L1 and M1 are available. It uses finite slopes and the support-wise MCA normalization used in this packet.

Proposition 1 (list/MCA bridge to line decoding)**.** *Let $C = \mathrm{RS}[\mathbb{F}, D, k]$, let $|D| = n$, and let $a = \lceil(1-\delta)n\rceil > k$. Suppose that every allowed received line $f + \gamma g$ has at most M support-wise MCA-bad finite slopes at radius δ, equivalently $\varepsilon_{\mathrm{mca}}(C, \delta) \leq M/|\mathbb{F}|$ for the chosen finite slope sampler. Suppose also that the two-row interleaved list size satisfies*

$$|\Lambda(C^{\equiv 2}, \delta, W)| \leq L \qquad \textit{for every } W \in (\mathbb{F}^2)^D.$$

Then C is line-decodable with parameters

$$(\delta,\ M + nL + 1,\ n + 1)$$

in the following sense: if an allowed received line $f + \gamma g$ has more than $M + nL$ finite slopes γ that are δ-close to selected codewords $U(\gamma) \in C$, then there are codewords $c_1, c_2 \in C$ such that

$$U(\gamma) = c_1 + \gamma c_2$$

for at least $n + 1$ of those slopes.

Proof. For each close slope γ, choose a support $S_\gamma \subseteq D$ of size a on which $f + \gamma g$ agrees with the selected codeword $U(\gamma)$. Discard the at most M support-wise MCA-bad slopes. For every remaining slope, support-wise containment gives codewords $c_{1,\gamma}, c_{2,\gamma} \in C$ such that

$$c_{1,\gamma}|_{S_\gamma} = f|_{S_\gamma}, \qquad c_{2,\gamma}|_{S_\gamma} = g|_{S_\gamma}.$$

Thus $(c_{1,\gamma}, c_{2,\gamma})$ is a two-row interleaved codeword δ-close to the received two-row word (f, g), using the common support S_γ. There are at most L possible pairs.

If the original line has more than $M + nL$ close slopes, then after discarding bad slopes more than nL slopes remain. By pigeonhole, one pair (c_1, c_2) occurs for at least $n + 1$ slopes. For each such slope, the codewords $U(\gamma)$ and $c_1 + \gamma c_2$ both agree with $f + \gamma g$ on S_γ, hence agree with each other on $a > k$ positions. Since both have degree less than k, they are identical codewords. This proves the stated line-decoding conclusion. ☐

Remark 2 (finite-constant consequence)**.** *If M1 gives $M = n^{1+o(1)}$ and L1 plus Lemma 1 gives $L = n^B$, then Proposition 1 gives*

$$M + nL + 1 \le n^{B+1+o(1)}.$$

For $n \le 2^{44}$ and $|\mathbb{F}| < 2^{256}$, a raw list numerator n^B is compatible with a 2^{-128} field budget only for about $B \lesssim 2.9$, and the line-decoding bridge is compatible only for about $B \lesssim 1.9$. Thus the bridge is useful, but it leaves a real finite-constant target.

4 Updated Positive-Theorem Package

The clean theorem to aim for is the following finite-reserve package. Let

$$C = \mathrm{RS}[\mathbb{F}_{\mathrm{line}}, D, k], \qquad n = |D|, \qquad a = k + \sigma, \qquad \delta = 1 - a/n = 1 - \rho - \eta.$$

Assume

$$\sigma \log_2 q_{\mathrm{gen}} \ge (1+\varepsilon) \log_2 \binom{n}{k+\sigma}, \qquad \sigma \ge C_{\rho,B,\varepsilon} \frac{n}{\log n},$$

the quotient profile $\mathcal{Q}_H(\eta)$ is empty or explicitly budgeted, η lies above all proved failure floors including Paper D's universal-cap floor, and the MCA or line experiment is proved over q_{line}, not merely over q_{chal}.

Conjecture 1 (corrected finite reserve package)**.** *Under the preceding hypotheses, the desired conclusions are*

$$|\Lambda(C^{\equiv\mu}, \delta)| \le n^B$$

for all ordinary column-distance protocol arities μ, by Lemma 1; and

$$\varepsilon_{\mathrm{mca}}(C, \delta) \le \frac{n^{1+o(1)} + 2^{(\beta(\rho)/H_2(\rho))\mathcal{Q}_H(\eta)(1+o(1))}}{q_{\mathrm{line}}},$$

or the same bound over q_{gen} plus a proved extension-line lift. If the consumed primitive is line decoding, then Proposition 1 should give

$$C \text{ is } (\delta,\ M + nL + 1,\ n+1) \text{ line-decodable},$$

with M and L supplied by M1 and L1.

This package would turn the current negative cap into a sharp threshold program: below the cap one cannot hope for 2^{-128} MCA soundness, while above the corrected reserve one would have finite numerators that can be checked against q_{line}.

5 Immediate Experimental Additions

1. Add `experimental/interleaved_fiber_injection.md`, containing Lemma 1. Update certificate logic so an L1 locator-fiber theorem gives $L_\mu = n^B$, not $n^{\mu B}$, for ordinary column-distance RS interleaving.

2. Add `experimental/list_mca_to_linedecoding.md`, containing Proposition 1. This gives a concrete M2 bridge and turns line decoding into a finite-constant ledger problem.

3. Build a certificate scanner reporting generated-field entropy margin, quotient profile after $k = \rho n - r$, Paper D cap floor, base and interleaved list numerators, MCA numerator, line-decoding numerator, field separation among q_{gen}, q_{line}, and q_{chal}, extension status, and exact failure audit.
4. Keep the Crites–Stewart import marked CONDITIONAL until audited. Paper D's cap is powerful enough to guide threshold claims, but the conversion is load-bearing.

6 Fresh Non-Claims

This revision does not claim an exact settlement of the smooth-case Proximity Prize threshold. It claims that the shape is now rigid:

$$\boxed{\text{No unslacked capacity theorem; prove a reserve theorem at } 1-\rho-\eta.}$$

The theorem-backed negative side gives a conditional universal cap at constant gap 2^{-9} for $\rho = 1/2, 1/4, 1/8$ and 2^{-10} for $\rho = 1/16$. The positive side should be pursued as

$$\text{L1 locator local limit} + \text{M1 residue-line packing} + \text{extension-line theorem} + \text{protocol-ledger rewrite.}$$

The interleaved-list problem is less fundamental than it looked in the old file: for column-distance interleaving, a uniform locator-fiber theorem immediately bounds the interleaved list at the same exponent.

7 Retained Previous Experiments Material

The remaining sections are retained from the previous `experiments.tex` whenever they still provide useful provenance, notation, normal forms, finite certificate gates, or non-claims. The new front matter above should be read as the current threshold-level guide; the older material below remains a toolbox.

8 Status, Provenance, and What Counts as New

This file is the place where we collect new things, but it must separate new repository contributions from imported constructions. The status labels used below are:

IMPORTED	an external theorem, proof pattern, or conversion;
WRAPPER	a repository repackaging, dictionary, or pigeonhole consequence;
NEW-LOCAL	a restricted algebraic theorem proved in this note;
TARGET	a precise proof obligation not yet promoted;
HEURISTIC	evidence or a search direction, not a theorem.

Hard attribution rule. Any locator-polynomial argument derived from the subgroup construction of Ben-Sasson–Carmon–Haböck–Kopparty–Saraf must cite [2, Thm. 7.1] where the construction is stated or proved. Any admissible subgroup instantiation must also cite [2, Thm. 1.13]. The citation belongs at the theorem statement and again at the proof entry point; a bibliography entry or acknowledgement alone is not enough.

What is new in this note. The following items are the repository-side contributions collected here.

- A smooth-quotient notation dictionary translating BCHKS subgroup notation into the RS-MCA notation $Q = D^{c_{\text{fib}}}$ and the dyadic/FFT-domain ledgers used in Papers A–D.
- A list-fiber pigeonhole wrapper: once a locator construction assigns many subset certificates to a smaller slope set, one heavy slope has many locator certificates, and, after a distinctness check, many nearby codewords.
- A slack-two and subfield-pigeonhole proof target for the Paper D route: the imported locator construction should be run at the augmented-code rung needed by the Crites–Stewart list-to-agreement conversion, and slopes should be pigeonholed over the generated/base field when the construction really takes values there.
- A New-local divisibility gate in the restricted Paper B window $B = \mathbb{F}_p$, $F = \mathbb{F}_{p^2}$, $D = \mathbb{F}_p$, $t = \sigma = 2$, $j = 3$. This is independent of the BCHKS priority issue and is the main theorem proved in Section 14.

What is not new. The polynomial identity behind the item-(d) locator proof is not new here. In BCHKS notation it is the expansion of a vanishing polynomial over a complement of a chosen subset; in smooth-quotient notation it is the same expansion of $\prod_{b \in A}(X^{c_{\text{fib}}} - b)$, up to choosing a subset or its complement. The use of ℓ for the chosen subset size also follows the external presentation closely enough that it should either be cited or renamed to ℓ_{pg}.

Source map. The main papers remain the authoritative documents.

- Paper A, `tex/RS_disproof_v3.tex`, refutes the old no-slack smooth-domain support-wise line-MCA statement. If it uses the locator construction, it should cite BCHKS and call the repository contribution a smooth-domain/list-fiber corollary, not a new locator proof.
- Paper B, `tex/slackMCA_v3.tex`, is the main target for the local resonance material in Sections 11–17.
- Paper C, `tex/snarks_v4.tex`, records protocol ledgers. Nothing in this note may be consumed by a protocol without a separate conversion theorem.
- Paper D, `tex/cs25_cap_v4.tex`, gives the universal cap via an imported list-to-agreement route. Its locator-side input should cite BCHKS explicitly if it uses the quotient-locator construction.

9 BCHKS Locator Construction in Smooth-Quotient Notation

Status: Imported plus Wrapper. This section is a provenance-safe restatement and dictionary. It is included so later main-paper edits have a clean place to copy from.

9.1 Imported theorem/proof pattern

BCHKS prove a general multiplicative-subgroup construction [2, Thm. 7.1]. In a compact form, the imported statement is as follows.

Theorem 1 (BCHKS quotient-locator construction, imported)**.** *Let $G \subseteq H \subseteq \mathbb{F}_q^*$ be multiplicative subgroups and let $\Phi : H \to G$ be $x \mapsto x^c$, where $c = |H|/|G|$. Let $E \subseteq G$, and let $\ell_{\mathrm{pg}}, a \geq 1$ satisfy*

$$|E^{(+\ell_{\mathrm{pg}})}| = \#\{e_1 + \cdots + e_{\ell_{\mathrm{pg}}} : e_i \in E \textit{ distinct}\} \geq a.$$

Let $D = \Phi^{-1}(E)$, $n = |D|$, and

$$C = \mathrm{RS}[\mathbb{F}_q, D, n - (\ell_{\mathrm{pg}} + 2)c].$$

Then, for $\gamma = \ell_{\mathrm{pg}} c/n$, there are functions $f, g : D \to \mathbb{F}_q$ such that

$$\#\{z \in \mathbb{F}_q : \Delta(f + zg, C) \leq \gamma\} \geq a,$$

while

$$\Delta([f, g], C^2) \geq (\ell_{\mathrm{pg}} + 1)c/n.$$

Reference. This is exactly the imported construction of [2, Thm. 7.1]; the proof is not reproduced as a repository contribution. BCHKS define the line endpoints by the monomials $x^{n-\ell_{\mathrm{pg}}c}$ and $x^{n-(\ell_{\mathrm{pg}}+1)c}$, then expand a vanishing polynomial over the complement of a chosen ℓ_{pg}-subset. The coefficient of $X^{n-(\ell_{\mathrm{pg}}+1)c}$ is an affine transform of a restricted sum from $E^{(+\ell_{\mathrm{pg}})}$, giving the exceptional slope values. □

The BCHKS Theorem 1.13 is the admissible-subgroup specialization of this construction: take $E = G$, take $H = D$, set $c = |D|/|G|$, and use $\ell_{\mathrm{pg}} = |G|/2$. Any RS-MCA statement corresponding to that specialization should cite [2, Thm. 1.13] directly.

9.2 Notation dictionary

BCHKS notation	RS-MCA notation	Meaning
H	evaluation domain D	Large multiplicative domain on which codewords are evaluated.
G	quotient $Q = D^{c_{\mathrm{fib}}}$	Image of the power map; one point in Q has c_{fib} preimages in D.
$c = \lvert H\rvert/\lvert G\rvert$	$c_{\mathrm{fib}} = n/\lvert Q\rvert$	Fiber size of the quotient map. Avoid calling this a, because BCHKS also uses a for the number of exceptional slopes.
$\Phi(x) = x^c$	$x \mapsto x^{c_{\mathrm{fib}}}$	Quotient/fiber map.
$E \subseteq G$	active quotient subset $E \subseteq Q$	Quotient-level support used by the locator.
$E^{(+\ell_{\mathrm{pg}})}$	restricted first-coefficient/slope set	Sums of distinct chosen quotient elements; slopes are affine transforms of these sums.
ℓ_{pg}	ℓ_{pg} or renamed subset size	Number of chosen quotient elements. Use ℓ_{pg} in RS-MCA text to avoid collision with other ℓ's.
$f = x^{n-\ell_{\mathrm{pg}}c}$, $g = x^{n-(\ell_{\mathrm{pg}}+1)c}$	locator line endpoints	Same monomial line after quotient notation is substituted.
$P_S(X) = \prod_{\alpha\in S'}(X^c - \alpha)$	$\prod_{b\in A}(X^{c_{\mathrm{fib}}} - b)$, possibly after complementing A	The locator polynomial.
exceptional z's	bad slopes / heavy list fibers	Slope parameters for which a line word is close to the RS code.

9.3 The shared locator identity

The following identity is written down because it is the exact place where priority can be lost. The identity is the BCHKS identity in RS-MCA notation.

Lemma 2 (Locator expansion, imported identity)**.** *Let D map c_{fib}-to-one onto a quotient set E by $x \mapsto x^{c_{\mathrm{fib}}}$, and let $A \subseteq E$ have size ℓ_{pg}. Put $A^c = E \setminus A$ and define*

$$P_A(X) = \prod_{\alpha\in A^c} (X^{c_{\mathrm{fib}}} - \alpha).$$

Then

$$P_A(X) = X^{n-\ell_{\mathrm{pg}}c_{\mathrm{fib}}} - e_1(A^c)X^{n-(\ell_{\mathrm{pg}}+1)c_{\mathrm{fib}}} + \textit{terms of degree at most } n - (\ell_{\mathrm{pg}} + 2)c_{\mathrm{fib}}.$$

Equivalently, since $e_1(A^c) = e_1(E) - e_1(A)$, the coefficient of the next monomial is an affine transform of the restricted sum $e_1(A)$.

Reference. This is the elementary expansion used inside [2, Thm. 7.1]. We record it only to fix notation. It should not be cited as a new lemma of the repository unless the BCHKS provenance is present at the statement. □

9.4 List-fiber pigeonhole wrapper

The next proposition is the repository-side wrapper around the imported identity. It is intentionally stated in certificate form, because distinctness of the nearby codewords is a separate check in concrete instantiations.

Proposition 2 (List-fiber pigeonhole wrapper)**.** *Let $\mathcal{A}$ be a family of ℓ_{pg}-subsets of a quotient set E. Suppose the BCHKS locator construction assigns to each $A \in \mathcal{A}$ a slope $z(A)$ lying in a finite set S, and a locator certificate showing that the corresponding word $f + z(A)g$ is within radius γ of a locator codeword $P_A \in C$. Then some slope $z_0 \in S$ has at least*

$$\frac{|\mathcal{A}|}{|S|}$$

locator certificates above it. If the codewords P_A attached to the certificates over z_0 are distinct, then the Hamming ball of radius γ around $f + z_0 g$ contains at least $|\mathcal{A}|/|S|$ codewords of C.

Proof. The first assertion is the pigeonhole principle applied to the map $A \mapsto z(A)$. The second assertion is only the definition of list size, after the distinctness check removes possible duplicate locator codewords. The construction that supplies the certificates is imported from Theorem 1. □

Remark 3 (What the wrapper adds)**.** *The new part here is not the locator polynomial. The wrapper isolates the extra bookkeeping needed by RS-MCA: which slope set pays the pigeonhole bill, whether the slope set is the ambient field $\mathbb{F}$ or a generated/base subfield B, and whether locator certificates remain distinct after specialization. Those are the checks needed before using the construction in Papers A, B, or D.*

9.5 Slack-two and subfield target for Paper D

Status: Target. The Paper D route needs a locator-heavy list fiber in the augmented code rung $C^+ = \mathrm{RS}[\mathbb{F}, D, k+1]$, because that is the object fed into the Crites–Stewart list-to-agreement conversion. The proof obligation should be stated as follows.

Proposition 3 (Slack-two/subfield locator target)**.** *Let $D \subseteq B \subseteq \mathbb{F}$, and suppose a BCHKS-type quotient locator construction over D produces slopes in B, not merely in $\mathbb{F}$. If the one-rung/augmented-code version produces distinct locator codewords in $C^+ = \mathrm{RS}[\mathbb{F}, D, k+1]$ for a family $\mathcal{A}$ of certificates, then there exists a slope $z_0 \in B$ with list size at least*

$$\frac{|\mathcal{A}|}{|B|}$$

inside the appropriate Hamming ball for C^+.

Bookkeeping only. This is Proposition 2 with $S = B$. The nontrivial work is external or elsewhere: one must verify the BCHKS locator hypotheses at the augmented-code rung, verify that the slope coefficients lie in B, and verify distinctness of the produced codewords. Until those checks are present, this proposition is a target, not a promoted Paper D theorem. □

9.6 Pasteable main-paper wording

Paper A wording. The polynomial locator identity used here is the quotient-fiber form of the construction of Ben-Sasson–Carmon–Haböck–Kopparty–Saraf [2, Thm. 7.1]; their Theorem 1.13 is the admissible subgroup instantiation. Thus the locator proof itself is not a new contribution of this manuscript. The present use is the smooth-domain specialization, the field/divisor bookkeeping, and the list-fiber pigeonhole packaging.

Paper D wording. The locator-side input is the BCHKS quotient-locator construction [2, Thm. 7.1; cf. Thm. 1.13], written in the smooth-domain quotient notation $Q = D^{c_{\mathrm{fib}}}$. The additional bookkeeping needed here is the slack-two augmented-code rung and, when available, subfield pigeonholing over B rather than over the ambient extension field $\mathbb{F}$.

10 Triage of the AI-Generated Four-Item Packet

Status: Audit. The four advertised items should be positioned as follows before any promotion into a main paper. The original AI packet used the labels (a)–(d) without making them self-contained in this note, so we decode them here.

Item	Short name	Meaning in this ledger
(a)	Weak-slack positive regime	A positive-looking proximity/list or MCA-adjacent statement that only reaches slack on the order of $1/\sqrt{n}$. It is useful as a comparison theorem, but it is far from the constant/reserve bookkeeping needed for the corrected RS-MCA target.
(b)	Finite Fermat-prime packet	A special finite-domain instantiation over only three Fermat-prime cases, in the current experimental ledger represented by the $p \in \{17, 257, 65537\}$ style checks. It is evidence or a worked computation, not a uniform theorem.
(c)	Exponential-field construction	A large-field variant whose proof requires the ambient field to be exponential in the domain/parameter size. This can clarify the mechanism but does not match the polynomial-field or challenge-field regimes needed by the Proximity Prize ledger.
(d)	BCHKS quotient-locator packet	The interesting locator/proximity-gap construction: choose a quotient of a multiplicative subgroup, use ℓ_{pg}-subsets and restricted sums to create many exceptional slopes or nearby locator certificates. This is the BCHKS construction of [2, Thm. 7.1], with the admissible subgroup instance in [2, Thm. 1.13]; the repository contribution is only the smooth-domain dictionary, list-fiber wrapper, and later slack-two/subfield bookkeeping.

Item (d) should not be confused with the New-local divisibility-gate theorem in Section 14. The former is an imported locator/proximity construction; the latter is an independent restricted Paper B resonance calculation.

Item	Limitation	Repository handling
(a)	Works only at slack on the order of $1/\sqrt{n}$.	Use as a weak-slack comparison point only. It does not clear the corrected-reserve MCA objective.
(b)	Applies only to three Fermat-prime cases.	Treat as finite-prime evidence or a worked computation, not as a uniform smooth-domain theorem.
(c)	Requires exponential field size.	Treat as a large-field sanity check. It is not directly usable for polynomial-field or prize-ledger parameters.
(d)	Most relevant, but the locator proof is essentially BCHKS.	Import, cite, and adapt from [2, Thm. 7.1 and Thm. 1.13]. The repository contribution is only the smooth-domain/list-fiber/slack-two packaging after exact hypotheses are checked.

Promotion checklist. Before any statement using item (d) moves to Papers A–D, check the following.

- Does the statement cite BCHKS at the theorem and proof entry point?
- Is the object a proximity-gap statement, a list statement, CA, MCA, line-decoding, or a protocol-facing theorem?
- Which field pays the slope/list pigeonhole bill: B, $\mathbb{F}$, or another generated field?
- Are the locator codewords distinct after specialization?
- Is the claimed slack in the same normalization as the consumed theorem?
- Are items (a), (b), and (c) kept out of asymptotic/protocol claims unless their limitations are explicitly part of the theorem statement?

11 Restricted Resonance Setup for Paper B

Status: New-local for the divisibility gate; otherwise local normal-form bookkeeping. The remaining sections preserve and reorganize the Cycle 14–18 algebraic material. This branch is independent of the BCHKS priority correction.

The common restricted setting is

$$B = \mathbb{F}_p, \qquad F = \mathbb{F}_{p^2}, \qquad D = \mathbb{F}_p, \qquad t = \sigma = 2, \qquad j = 3,$$

and we work off the tangent/global endpoint

$$R_0 = \{\, [W]_E \wedge [B_{\mathrm{num}}]_E = 0 \,\}.$$

The field ledgers remain distinct:

$$q_{\mathrm{gen}} = p, \qquad q_{\mathrm{line}} = p^2.$$

The regime is sub-reserve, since $\eta = \sigma/n = 2/n$. Thus none of the following proves a corrected-reserve list, CA, MCA, line-decoding, SNARK, or protocol theorem.

The base/generated field is $B = \mathbb{F}_p$. The line field is the quadratic extension $F = \mathbb{F}_{p^2} = B(\alpha)$, with basis $\{1, \alpha\}$ over B and $\alpha^2 = \nu \in B$. The domain is $D = \mathbb{F}_p$, so $n = |D| = p$.

A degree-two residue-line datum consists of

$$E \in F[X], \qquad \deg E = 2, \qquad E(x) \neq 0 \text{ for all } x \in D,$$

and a numerator

$$B_{\text{num}} \in F[X], \qquad \deg B_{\text{num}} < 2.$$

We write

$$A = F[X]/E, \qquad [P]_E \in A, \qquad b = [B_{\text{num}}]_E.$$

The anchor word on D is interpolated by a polynomial W. The operation $u \wedge v$ means the determinant of the coordinate vectors of $u, v \in A$ in a chosen F-basis of the two-dimensional F-space A. Off R_0, the pair $\{[W]_E, b\}$ is such a basis.

The co-support has size $j = 3$. If $T \subset D$ is the three-point co-support, define

$$\tau_1 = e_1(T), \qquad \tau_2 = e_2(T), \qquad \tau_3 = e_3(T).$$

All valid co-supports have $\tau_i \in B$. The quantity C_2 denotes the number of distinct slopes produced by the relevant landing branch after restricting to valid split triples $T \subset \mathbb{F}_p$. Bounding landing points therefore bounds C_2, but a sharp slope theorem may require more because many landing points can share a slope.

12 Cycle 14 Normal Form

The Cycle 14 audit reduces the $t = 2, j = 3$ landing equation to two affine forms in the elementary co-support parameters

$$\tau = (\tau_1, \tau_2, \tau_3).$$

Lemma 3 (Affine wedge normal form)**.** *Assume $\{[W]_E, b\}$ is an F-basis of A. Then the landing equation can be written in the form*

$$\iota(\tau) = A_0(\tau_1, \tau_2) - \tau_3 [W]_E, \qquad \mu(\tau) = B_0(\tau_1, \tau_2) - \tau_3 b,$$

where

$$A_0 = p_1 [W]_E + p_2 b, \qquad B_0 = q_1 [W]_E + q_2 b,$$

and $p_i, q_i \in F$ are affine-linear functions of (τ_1, τ_2). If the basis is normalized so $[W]_E \wedge b = 1$, then

$$\Delta = \iota \wedge \mu = (p_1 - \tau_3)(q_2 - \tau_3) - p_2 q_1.$$

Proof. The Cycle 14 quotient calculation, which is a concrete instance of Paper B's residue-line normal form, gives the displayed affine forms. Expanding in the ordered basis $\{[W]_E, b\}$, the coordinate vector of ι is $(p_1 - \tau_3, p_2)$, while the coordinate vector of μ is $(q_1, q_2 - \tau_3)$. Their wedge determinant is therefore

$$(p_1 - \tau_3)(q_2 - \tau_3) - p_2 q_1.$$

□

13 Cycle 18 Component Split

Choose a basis $F = B + \alpha B$ with $\alpha^2 = \nu \in B$, and write

$$p_1 = p_{10} + \alpha p_{11}, \quad p_2 = p_{20} + \alpha p_{21}, \quad q_1 = q_{10} + \alpha q_{11}, \quad q_2 = q_{20} + \alpha q_{21}.$$

Split

$$\Delta = \Delta_0 + \alpha\Delta_1, \qquad \Delta_0, \Delta_1 \in B[\tau_1, \tau_2, \tau_3].$$

Theorem 2 (Monic quadratic and linear alpha component)**.** *In the setting above,*

$$\begin{aligned}\Delta_0 &= \tau_3^2 - (p_{10} + q_{20})\tau_3 + p_{10}q_{20} + \nu p_{11}q_{21} - p_{20}q_{10} - \nu p_{21}q_{11},\\ \Delta_1 &= \ -(p_{11} + q_{21})\tau_3 + p_{10}q_{21} + p_{11}q_{20} - p_{20}q_{11} - p_{21}q_{10}.\end{aligned}$$

Consequently Δ_0 is monic quadratic in τ_3, while Δ_1 has degree at most one in τ_3.

Proof. Substitute the coordinate expressions for p_i, q_i into the identity of Lemma 3 and use $\alpha^2 = \nu$. Collecting the coefficient of 1 gives the displayed formula for Δ_0, and collecting the coefficient of α gives the displayed formula for Δ_1. The coefficient of τ_3^2 in Δ_0 is 1, and the α-component has no τ_3^2-term. □

Corollary 1 (Graph reduction outside the zero-alpha branch)**.** *If Δ_1 is not the zero polynomial, write*

$$\Delta_1 = s(\tau_1, \tau_2)\tau_3 + h(\tau_1, \tau_2).$$

On the open locus $s \neq 0$, the common-zero condition $\Delta_0 = \Delta_1 = 0$ is supported on the graph

$$\tau_3 = -h/s.$$

Proof. This is immediate from solving the linear equation $\Delta_1 = 0$ for τ_3 on $s \neq 0$. □

14 New Divisibility-Gate Theorem

The next theorem is the main New-local theorem extracted from the Cycle 18 reduction. It is stronger than merely saying that the branch is a graph: it shows that any two-dimensional family must satisfy an explicit divisibility identity.

Theorem 3 (Divisibility gate for the graph branch)**.** *Assume the restricted setting above and assume $\Delta_1 \neq 0$. Write*

$$\Delta_1 = s(\tau_1, \tau_2)\tau_3 + h(\tau_1, \tau_2),$$

and define the denominator-cleared graph polynomial

$$G(\tau_1, \tau_2) = s(\tau_1, \tau_2)^2 \Delta_0(\tau_1, \tau_2, -h(\tau_1, \tau_2)/s(\tau_1, \tau_2)).$$

Then $\deg G \leq 4$. If G is not the zero polynomial, then

$$\#\{\, \tau \in B^3 : \Delta_0(\tau) = \Delta_1(\tau) = 0 \,\} = O(p).$$

Consequently the corresponding number of distinct slopes satisfies

$$C_2 = O(p) = O(n).$$

Therefore a growing-prime family with

$$C_2 = \Theta(p^2) = \Theta(q_{\text{line}})$$

in the nonzero-Δ_1 branch can occur only if $G \equiv 0$.

Proof. Because the p_i, q_i are affine-linear in (τ_1, τ_2), the coefficient s has degree at most 1, and h has degree at most 2. Since Δ_0 is monic quadratic in τ_3, clearing the denominators after substituting $\tau_3 = -h/s$ gives a polynomial G of degree at most 4.

On $s \neq 0$, the equation $\Delta_1 = 0$ forces $\tau_3 = -h/s$. Hence every common zero on this open locus gives a base pair (τ_1, τ_2) with $G(\tau_1, \tau_2) = 0$. If G is nonzero, Schwartz–Zippel gives at most $4p$ such base pairs, and each base pair gives at most one value of τ_3.

It remains to count the exceptional locus $s = 0$. There $\Delta_1 = 0$ also forces $h = 0$. If s is a nonzero affine form, this is contained in an affine line; if $s \equiv 0$, then h is a nonzero polynomial of degree at most 2. In either case there are $O(p)$ base pairs. For each such pair, Δ_0 is a monic quadratic in τ_3, so it gives at most two values of τ_3. The total common-zero count is $O(p)$.

Finally, the number of distinct slopes is bounded by the number of landing points τ. Thus $C_2 = O(p)$ whenever $G \not\equiv 0$, and any $\Theta(p^2)$ family in this branch must satisfy $G \equiv 0$. ☐

Remark 4. *The identity $G \equiv 0$ says that, after clearing the s-denominator, Δ_0 is divisible by the linear equation Δ_1 on the graph branch. This is the first exact algebraic gate that a large slope family in this restricted window must pass.*

15 Nearby Proven Gates

The same restricted lane contains two other useful gates. They are included here because they explain where Theorem 3 fits.

Proposition 4 (Base-component complete-intersection gate)**.** *Suppose $\Delta_0, \Delta_1 \in B[\tau_1, \tau_2, \tau_3]$ have no common surface component. Then*

$$\#V(\Delta_0, \Delta_1)(B) = O(p),$$

and hence this branch contributes $C_2 = O(p)$ slopes.

Proof. The two components have bounded degree. If they share no surface component, their common zero set in $\mathbb{A}^3_B$ is a curve of bounded degree. A bounded-degree affine curve over $\mathbb{F}_p$ has $O(p)$ rational points, and slopes are bounded by landing points. ☐

Proposition 5 (Cycle 16 determinant gate)**.** *Let $Q(z_0, z_1)$ be the Cycle 15–16 determinant consistency polynomial for the affine system*

$$L_z(\tau) = \iota(\tau) - z\mu(\tau) = 0, \qquad z = z_0 + \alpha z_1.$$

If Q is not identically zero, then in the restricted $D = \mathbb{F}_p$, $t = \sigma = 2$, $j = 3$, off-R_0 window,

$$C_2 \leq 4p = O(p) = O(n).$$

Proof. Cycle 16 gives $\deg Q \leq 4$ and shows that every landing slope must satisfy $Q(z_0, z_1) = 0$. If $Q \neq 0$, Schwartz–Zippel over $B = \mathbb{F}_p$ gives at most $4p$ possible pairs (z_0, z_1), hence at most $4p$ slopes. ☐

16 Slope Map and Remaining Heuristics

On the graph branch, the landing equation $\iota = z\mu$ gives

$$p_1 - \tau_3 = zq_1, \qquad p_2 = z(q_2 - \tau_3).$$

Where $q_1 \neq 0$, substituting $\tau_3 = -h/s$ yields

$$z(\tau_1, \tau_2) = \frac{p_1 + h/s}{q_1}.$$

Equivalently, eliminating τ_3 recovers the Cycle 14 quadratic

$$q_1 z^2 - (p_1 - q_2)z - p_2 = 0,$$

so the residual map is controlled by

$$(\tau_1, \tau_2) \longmapsto [\, q_1 : (p_1 - q_2) : p_2 \,] \in \mathbb{P}^2(F).$$

Heuristic 1 (Exceptional identity should be rare)**.** *For generic source data, the polynomial G should not vanish identically. Thus the graph branch should usually be curve-sized, and the first possible large-slope families should come from algebraically special data satisfying the divisibility identity $G \equiv 0$ or from the zero-alpha branch $\Delta_1 \equiv 0$.*

Heuristic 2 (What remains after the gate)**.** *If $G \equiv 0$, the problem is no longer to show that the common-zero set is small. It may be surface-sized. The remaining question is whether the projective coefficient map has one-dimensional image on every source-valid surface, giving $C_2 = O(p)$, or whether some growing-prime source-valid family has two-dimensional image and $C_2 = \Theta(p^2)$.*

17 Next Experimental Checks

A useful scanner should compute, for each source-valid candidate,

$$\Delta_1 = s\tau_3 + h, \qquad G = s^2 \Delta_0(\tau_1, \tau_2, -h/s),$$

then record whether $G \equiv 0$, whether $\Delta_1 \equiv 0$, the exceptional locus $s = h = 0$, the image size of

$$[\, q_1 : (p_1 - q_2) : p_2 \,],$$

and the direct slope count over distinct split triples $T \subset \mathbb{F}_p$.

The certificate should keep at least

$$p,\ q_{\text{gen}},\ q_{\text{line}},\ E,\ B_{\text{num}},\ \text{seed},\ \text{stratum},\ R_0\text{-status},$$
$$\Delta_1 \equiv 0, \quad G \equiv 0, \quad \#\{s = h = 0\}, \quad \#\,\text{image}, \quad C_2, \quad \#\{T \text{ split distinct}\}.$$

18 June 26 Normal-Form and Dependency Results

Status: mixed. This section records the parts of the June 26 PR batch that survived a statement-level validity check. The statements below are useful proof infrastructure, not prize claims. In particular, they do not improve the current Cycle116/119 numerator 52,747,567,092 over $\mathbb{F}_{17^{32}}$, and they do not prove the corrected Paper B MCA conjecture.

18.1 Syndrome-pencil normal form for support-wise line incidence

Status: New-local. This is the cleanest reusable F1 contribution from the batch. It turns support-wise line incidence into a Hankel-pencil locator equation over the actual line field.

Let $D = \{x_1, \ldots, x_n\} \subset F$ have distinct points and let $C = \mathrm{RS}[F, D, k]$ with redundancy $r = n - k$. For each x_i, put

$$\lambda_i = \left(\prod_{h \neq i}(x_i - x_h)\right)^{-1}.$$

For a word $y : D \to F$, define

$$\mathrm{Syn}(y)_m = \sum_i \lambda_i x_i^m y(x_i), \qquad 0 \leq m < r.$$

Then $\mathrm{Syn}(y) = 0$ if and only if $y \in C$. Fix $T \subset D$ with $|T| = j$, assume $0 \leq j \leq r$, and set $t = r - j$. Write

$$L_T(X) = \prod_{x \in T}(X - x) = \ell_0 + \ell_1 X + \cdots + \ell_j X^j, \qquad \boldsymbol{\ell}_T = (\ell_0, \ldots, \ell_j)^T.$$

For $w \in F^r$, define the $t \times (j+1)$ Hankel window

$$H_{t,j}(w)_{a,b} = w_{a+b}, \qquad 0 \leq a < t, \quad 0 \leq b \leq j.$$

Theorem 4 (Hankel-pencil test)**.** *Let $f, g : D \to F$, put $u = \mathrm{Syn}(f)$ and $v = \mathrm{Syn}(g)$, and let $S = D \setminus T$. For a finite slope $z \in F$, the word $f + zg$ is explained by a codeword of C on S if and only if*

$$(H_{t,j}(u) + zH_{t,j}(v))\boldsymbol{\ell}_T = 0.$$

Moreover, this explanation is support-wise noncontained for the line $f + zg$ on S if and only if, in addition,

$$H_{t,j}(v)\boldsymbol{\ell}_T \neq 0.$$

Consequently the support-wise MCA bad slopes at agreement $k + t$ are exactly the slopes for which there is a squarefree D-split locator L_T of degree $j = r - t$ satisfying the displayed incidence and noncontainment tests.

Proof. Let $W_T \subset F^r$ be the span of the parity-check columns indexed by T:

$$W_T = \mathrm{span}_F\{\lambda_x(1, x, \ldots, x^{r-1}) : x \in T\}.$$

Equivalently, $w \in W_T$ means that w is the syndrome of a word supported on T. For every $w \in W_T$, write

$$w_m = \sum_{x \in T} c_x x^m.$$

Then, for $0 \leq a < t$,

$$\left(H_{t,j}(w)\boldsymbol{\ell}_T\right)_a = \sum_{b=0}^{j} \ell_b w_{a+b} = \sum_{x \in T} c_x x^a L_T(x) = 0.$$

Conversely, since L_T is monic, the recurrence $\sum_{b=0}^{j} \ell_b w_{a+b} = 0$ for $0 \leq a < r - j$ determines all remaining coordinates from the first j coordinates, so its solution space has dimension at most j. The j columns indexed by the distinct points of T form a Vandermonde system of rank j, hence the recurrence space is exactly W_T.

The word $f + zg$ is explained on S if and only if there is $c \in C$ such that $f + zg - c$ is supported on T. Taking syndromes, this is equivalent to $u + zv \in W_T$, which is the first displayed equation by the recurrence criterion. Under that equation, simultaneous explanation of f and g on the same S is equivalent to $u, v \in W_T$. Since $u = (u + zv) - zv$, this is equivalent to $v \in W_T$, hence to $H_{t,j}(v)\boldsymbol{\ell}_T = 0$. Noncontainment is therefore exactly the negation of that condition. □

18.2 Interleaved codegree reduction

Status: New-local reduction; saving remains conditional. This result is useful for the L2 interleaved-list ledger because it replaces the naive Cartesian product by a recursion through higher-agreement tails.

Let $C = \mathrm{RS}[F, D, k]$. For a word $V : D \to F$ and codeword $c \in C$, write

$$A_V(c) = \{x \in D : c(x) = V(x)\}, \qquad \mathrm{Fib}_V(s) = \{c \in C : |A_V(c)| \geq s\},$$

and $M_V(s) = |\,\mathrm{Fib}_V(s)|$. For a μ-row word $U = (U_1, \dots, U_\mu)$, define

$$\Lambda_\mu^{(s)}(U) = \#\{(c_1, \dots, c_\mu) \in C^\mu : |\cap_i A_{U_i}(c_i)| \geq s\}.$$

Theorem 5 (two-regime codegree bound)**.** *For every agreement threshold a,*

$$\Lambda_2^{(a)}(U_1, U_2) \leq M_{U_2}(a) + M_{U_2}(2a - k)\, M_{U_1}(a).$$

More generally,

$$\Lambda_\mu^{(a)}(U_1, \dots, U_\mu) \leq \Lambda_{\mu-1}^{(a)}(U_2, \dots, U_\mu) + \Lambda_{\mu-1}^{(2a-k)}(U_2, \dots, U_\mu)\, M_{U_1}(a).$$

Proof. For the two-row case, sum over $c_2 \in \mathrm{Fib}_{U_2}(a)$, and put $N_2 = |A_{U_2}(c_2)|$. The possible c_1's are codewords agreeing with U_1 on at least a points of $A_{U_2}(c_2)$. If $N_2 < 2a - k$, then two such c_1's would agree with each other on more than k points, forcing equality. Thus this punctured list has size at most one. If $N_2 \geq 2a - k$, use the trivial bound $M_{U_1}(a)$. Summing these two regimes gives the displayed two-row inequality.

For the μ-row case, fix $(c_2, \dots, c_\mu)$ and put

$$S = \bigcap_{i=2}^{\mu} A_{U_i}(c_i).$$

The listed c_1's must agree with U_1 on at least a points of S. If $|S| < 2a - k$, the same unique-decoding argument gives at most one such c_1. If $|S| \geq 2a - k$, use the bound $M_{U_1}(a)$. Counting the two classes of $(c_2, \dots, c_\mu)$ gives the recursion. □

Remark 5. *The reduction is unconditional. The desired L2 saving still requires an L1-family input saying that the tail lists at agreement $2a - k$ are small after quotient-periodic contributions are budgeted.*

18.3 Deep-point cap dependency split

Status: Wrapper / Audit. This result does not edit Paper D, but it explains why the headline MCA cap can be routed locally without using the imported Crites–Stewart conversion.

Proposition 6 (CS25-free route to the Paper D MCA cap)**.** *Assume the finite-field, coset, divisibility, and binomial hypotheses used in Paper D's universal cap theorem. Let* $C = \mathrm{RS}[F, D, k]$, $C^+ = \mathrm{RS}[F, D, k+1]$, $q = |F|$*, and suppose the elementary quotient-locator fiber construction gives a* C^+*-list of size*

$$L \geq \binom{N}{\rho N + 2} / |B| \geq q/k + 1$$

at the endpoint $\delta_N = 1 - \rho - 2/N$*. Then there is a support-wise MCA-bad line for* C *with bad-slope density strictly larger than*

$$\frac{1}{2k}\left(1 - \frac{n}{q}\right).$$

By support-wise MCA monotonicity, the same lower bound holds for all $\delta_N \leq \delta < 1 - \rho$*.*

Proof. Let $U : D \to F$ be the received word with a C^+-list $\{P_1, \ldots, P_L\}$ at agreement $m = k+2n/N$. For $\alpha \in F \setminus D$, form

$$f_\alpha(x) = \frac{U(x)}{x - \alpha}, \qquad g_\alpha(x) = -\frac{1}{x - \alpha}.$$

For every listed P_i, the slope $z = P_i(\alpha)$ gives the codeword

$$Q_i(X) = \frac{P_i(X) - P_i(\alpha)}{X - \alpha} \in F[X]_{<k}$$

on the same agreement support. Conversely every such slope comes from a listed P_i. Since $m > k$, the direction g_α cannot be explained by a degree-$< k$ polynomial on the same support, so these slopes are support-wise MCA-bad.

Averaging over $\Omega = F \setminus D$, two distinct degree-$< k + 1$ polynomials collide at at most k points of Ω. Therefore some α has at least

$$M \geq \frac{L}{1 + k(L-1)/(q-n)}$$

distinct values among $P_1(\alpha), \ldots, P_L(\alpha)$. Its bad-slope density is at least

$$\frac{L(q-n)}{q(q - n + k(L-1))}.$$

This is strictly greater than $(q - n)/(2kq)$ exactly when

$$2kL > q - n + k(L - 1).$$

The lower bound $L \geq q/k+1$ gives $kL \geq q+k$, so the left-minus-right quantity is at least $n+2k > 0$. This proves the endpoint bound, and monotonicity in δ extends it to the stated interval. □

18.4 Common code-line residual budget

Status: New-local finite MDS theorem. This M2 lemma is a guardrail for line-decoding imports: a close common code-line exception still pays for residual coordinates outside the common support.

Proposition 7 (common code-line residual budget)**.** *Let $C \subseteq F^D$ be an MDS linear code of dimension k, so every nonzero codeword has fewer than k zeros on D. Let $\ell_z = f + zg$ be a received line and suppose that for some codewords $c_f, c_g \in C$ there is a common support $S_0 \subseteq D$ of size b such that $f = c_f$ and $g = c_g$ on S_0. Fix an agreement threshold a with $a + b - n \geq k$. Put $\Omega = D \setminus S_0$,*

$$f' = f - c_f, \qquad g' = g - c_g, \qquad h = \max(1, a - b),$$

and

$$c_0 = \#\{x \in \Omega : f'(x) = g'(x) = 0\}.$$

Every support-wise noncontained slope at agreement a satisfies

$$\#\{x \in \Omega : f'(x) + zg'(x) = 0\} \geq h.$$

Consequently, if $h > c_0$, then the number of such slopes is at most

$$\left\lfloor \frac{|\Omega| - c_0}{h - c_0} \right\rfloor .$$

Proof. If ℓ_z is explained on a support S of size at least a, then

$$|S \cap S_0| \geq a - |\Omega| = a + b - n \geq k.$$

On $S \cap S_0$, the explaining codeword agrees with $c_f + zc_g$. By the MDS zero-rigidity condition, the two codewords are equal. Thus, outside S_0, the line can only be explained where $f' + zg' = 0$, and at least $h = \max(1, a - b)$ such outside zeros are needed for a noncontained support.

Each non-common residual coordinate $x \in \Omega$ with $(f'(x), g'(x)) \neq (0, 0)$ can vanish for at most one finite slope z. A bad slope needs at least $h - c_0$ private non-common residual zeros after the c_0 common zeros are counted. Charging these private coordinates to slopes gives the stated packing bound. □

19 High-Agreement Tangent Staircase

Status: New-local finite MDS theorem. This section records a generic moving-root tangent floor and, in the very-high-agreement range, a matching upper bound. It is independent of quotient-periodic structure and does not use smoothness. It is relevant to the finite row ledger because the bad slopes are paid with the actual line field q_{line}.

For a polynomial $P \in F[X]$, write

$$P = P_{<k} + P_{\geq k}, \qquad \deg P_{<k} < k,$$

where $P_{\geq k}$ is the sum of the terms of degree at least k. Let $\mathrm{LD}_{\mathrm{sw}}(C, a)$ denote the largest possible number of finite slopes on a received line that have a support-wise noncontained explanation on some support of size at least a.

Theorem 6 (moving-root tangent floor)**.** *Let $D \subset F$ have $|D| = n$, and let $C = \mathrm{RS}[F, D, k]$. For every integer a with $k + 1 \leq a \leq n$,*

$$\mathrm{LD}_{\mathrm{sw}}(C, a) \geq n - a + 1.$$

Proof. Choose a subset $A \subset D$ of size $a-1$, and put

$$B_A(X) = \prod_{x \in A}(X - x).$$

Define the received line

$$f = (XB_A)_{\geq k}, \qquad g = -(B_A)_{\geq k}.$$

For every $t \in D \setminus A$, set

$$L_t(X) = (X - t)B_A(X).$$

Then L_t is the locator of the support $A \cup \{t\}$, which has size a, and linearity of truncation gives

$$f + tg = (L_t)_{\geq k}.$$

On every root of L_t, the word represented by $(L_t)_{\geq k}$ agrees with $-(L_t)_{<k}$, a codeword of C. Thus the slope t is explained on $A \cup \{t\}$.

It remains to check support-wise noncontainment on this same support. On A, since $B_A = 0$, we have

$$g = -(B_A)_{\geq k} = (B_A)_{<k}.$$

If g were explained on $A \cup \{t\}$ by a degree-$< k$ polynomial q, then $q = (B_A)_{<k}$, because $|A| = a - 1 \geq k$. At the remaining point t, this would force

$$-(B_A)_{\geq k}(t) = (B_A)_{<k}(t),$$

that is, $B_A(t) = 0$, contrary to $t \notin A$. Hence g is not separately explained on $A \cup \{t\}$, so the slope is support-wise noncontained. The slopes $t \in D \setminus A$ are distinct, giving $n - a + 1$ bad finite slopes. $\square$

Theorem 7 (very-high-agreement tangent staircase)**.** *Let $C = \mathrm{RS}[F, D, k]$ with $|D| = n$. If*

$$3a - 2n \geq k,$$

then

$$\mathrm{LD}_{\mathrm{sw}}(C, a) = n - a + 1.$$

Proof. The lower bound is Theorem 6. For the upper bound, put $s = n - a$. If a received line has at most one support-wise noncontained finite slope at agreement at least a, then the claimed upper bound is immediate. Otherwise pick two such slopes $z_1 \neq z_2$, with explaining codewords $p_1, p_2 \in C$ on supports S_1, S_2, respectively. Since $|S_i| \geq a$,

$$|S_1 \cap S_2| \geq 2a - n.$$

On this intersection,

$$g = (p_1 - p_2)/(z_1 - z_2), \qquad f = p_1 - z_1 g.$$

The condition $3a - 2n \geq k$ and the inequality $a \leq n$ imply $2a - n \geq k$. Therefore these two right-hand sides extend uniquely to codewords $c_g, c_f \in C$. Let S_0 be a maximal common support on which $f = c_f$ and $g = c_g$, and put $b = |S_0|$ and $d = n - b$. Then

$$b \geq 2a - n, \qquad d \leq 2(n - a) = 2s.$$

By maximality, outside S_0 there is no coordinate at which both residuals $f - c_f$ and $g - c_g$ vanish.

The residual-budget proposition, Proposition 7, is applicable because

$$a + b - n \geq a + (2a - n) - n = 3a - 2n \geq k.$$

Here its common-zero parameter is $c_0 = 0$. If $b \geq a$, then every noncontained slope needs at least one residual zero outside S_0, and there are only $d \leq s$ such coordinates. If $b < a$, then every noncontained slope needs at least

$$h = a - b = d - s$$

private residual zeros outside S_0. Since $s < d \leq 2s$, the residual budget gives

$$\#\{\text{bad slopes}\} \leq \left\lfloor \frac{d}{d-s} \right\rfloor \leq s + 1 = n - a + 1.$$

This proves the matching upper bound. □

Corollary 2 (tangent-star extremizers in the exact range)**.** *Let $C = \mathrm{RS}[F, D, k]$, $|D| = n$, and suppose that*

$$k + 1 \leq a \leq n, \qquad 3a - 2n \geq k.$$

If a received line $f + zg$ has exactly $n - a + 1$ support-wise noncontained finite bad slopes at agreement at least a, and if $n - a + 1 \geq 2$, then the line is tangent-star extremal in the following sense. There are codewords $c_f, c_g \in C$ and a common support $S_0 \subseteq D$ of size $a - 1$ such that, after replacing f, g by $f - c_f, g - c_g$, every bad slope is obtained by choosing exactly one residual coordinate in $D \setminus S_0$. More precisely, writing $\Omega = D \setminus S_0$, the map

$$x \longmapsto -\frac{f(x) - c_f(x)}{g(x) - c_g(x)}$$

is a bijection from Ω to the bad-slope set.

Proof. Choose two distinct bad slopes and form the maximal common code-line support S_0 as in the proof of Theorem 7; write $b = |S_0|$, $s = n - a$, and $d = n - b$. Equality in the upper bound from that proof cannot occur when $b \geq a$, because then there are at most $n - a$ bad slopes. Hence $b < a$. The residual-budget bound is

$$\#\{\text{bad slopes}\} \leq \left\lfloor \frac{d}{d-s} \right\rfloor \leq s + 1 = n - a + 1.$$

For equality, the second inequality must be sharp, so $d - s = 1$, equivalently $b = a - 1$. The maximality of S_0 gives no common residual zeros outside S_0. Since $|\Omega| = n - a + 1$ and each residual coordinate can vanish for at most one finite slope, equality forces every residual coordinate to be used exactly once. This is precisely the displayed residual-root bijection. □

Corollary 3 (exact high-agreement gate for the $\mathbb{F}_{17^{32}}$, rate-1/2 row)**.** *Let $H \subset \mathbb{F}_{17^{32}}^*$ be any multiplicative subgroup of order* 512*, and let*

$$C = \mathrm{RS}[\mathbb{F}_{17^{32}}, H, 256].$$

For every $a \geq 427$,

$$\mathrm{LD}_{\mathrm{sw}}(C, a) = 513 - a.$$

In particular,

$$\mathrm{LD}_{\mathrm{sw}}(C, 506) = 7, \qquad \mathrm{LD}_{\mathrm{sw}}(C, 507) = 6.$$

Since

$$6 \cdot 2^{128} < 17^{32} < 7 \cdot 2^{128},$$

we have

$$\frac{\mathrm{LD}_{\mathrm{sw}}(C, 507)}{17^{32}} < 2^{-128} \quad \textit{but} \quad \frac{\mathrm{LD}_{\mathrm{sw}}(C, 506)}{17^{32}} > 2^{-128}.$$

Thus, under the finite-slope support-wise MCA convention, the 2^{-128} *integer agreement staircase for this row is pinned between* 506 *and* 507*. The largest safe integer Hamming radius is* 5*, while integer radius* 6 *is already unsafe. The corresponding normalized grid radii are*

$$5/512 \quad \textit{and} \quad 6/512 = 3/256.$$

As a real-radius supremum, the transition is $3/256$*, but the closed endpoint* $\delta = 3/256$ *is on the unsafe side.*

Proof. Here $n = 512$ and $k = 256$. The hypothesis $3a - 2n \geq k$ becomes $3a - 1024 \geq 256$, hence $a \geq 427$. The exact formula is Theorem 7. The arithmetic inequalities are

$$17^{32} - 6 \cdot 2^{128} = 326217393234836465063858652730341978625 > 0$$

and

$$7 \cdot 2^{128} - 17^{32} = 140649736861019983995159547014262328310 > 0.$$

□

Corollary 4 (no seventh finite-slope branch past agreement 506)**.** *For* $C = \mathrm{RS}[\mathbb{F}_{17^{32}}, H, 256]$ *with* $|H| = 512$*, no finite-slope support-wise MCA mechanism can add a seventh bad slope at agreement* $a \geq 507$*. At equality, any extremal line is tangent-star in the sense of Corollary 2.*

Proof. By Corollary 3,

$$\mathrm{LD}_{\mathrm{sw}}(C, a) = 513 - a \qquad (a \geq 427).$$

For every $a \geq 507$, the right-hand side is at most 6. Thus a seventh finite support-wise bad slope is impossible in this range. The extremizer classification is Corollary 2. □

Remark 6 (effect on the earlier agreement-353 target)**.** *The agreement-353 target is no longer a frontier for the rate-*$1/2$*,* $\mathbb{F}_{17^{32}}$*,* $n = 512$ *row: Theorem 6 already gives*

$$\mathrm{LD}_{\mathrm{sw}}(C, 353) \geq 512 - 353 + 1 = 160.$$

The quotient-core strict352 packet remains a useful mechanism record, but the generic tangent floor gives a stronger lower bound at agreement 352 *and beyond. Corollary 4 rules out a seventh finite-slope branch past agreement* 506*. The adjacent-ledger section below treats the high-agreement projective-slope, CA, curve-MCA, and interleaved-list coding objects; protocol-facing conversions still require separate challenge-field, extension-lift, folding, query, and cryptographic ledgers.*

20 High-Agreement Adjacent Ledgers

Status: New-local finite Reed–Solomon theorem package, one MDS interleaved-list theorem, and a Conditional protocol-ledger corollary. The tangent staircase of Section 19 settles more than finite-slope support-wise MCA. In the same high-agreement range it also pins the no-loss correlated-agreement line count, the projective-slope line count, the finite-parameter degree-d curve count, and the interleaved-list size. The protocol statement at the end of this section is only a ledger corollary: it applies to reductions whose coding error is exactly one of the listed terms, plus an explicitly added query/folding error.

Except in the interleaved-list theorem, $C = \mathrm{RS}[F, D, k]$ with $|D| = n$, dimension k, and $q = |F|$. The interleaved-list uniqueness statement only uses the MDS property. The agreement parameter is an integer a, so the corresponding closed Hamming radius is $1 - a/n$.

20.1 No-loss CA and projective slopes

Let $\mathrm{LD}_{\mathrm{ca}}(C, a)$ denote the maximum number of finite slopes $z \in F$ on an affine received line $f + zg$ such that $f + zg$ has a C-explanation on at least a positions while the pair (f, g) has no C^2-explanation on any support of size at least a. This is the integer-agreement version of no-loss CA. Let $\mathrm{LD}_{\mathrm{sw,proj}}(C, a)$ be the corresponding support-wise MCA count when slopes are points of $\mathbb{P}^1(F)$, so the point at infinity is also allowed.

Theorem 8 (high-agreement CA and projective tangent staircase)**.** *Let $C = \mathrm{RS}[F, D, k]$ with $|D| = n$. If*

$$3a - 2n \geq k,$$

then

$$\mathrm{LD}_{\mathrm{ca}}(C, a) = \mathrm{LD}_{\mathrm{sw,proj}}(C, a) = n - a + 1.$$

Proof. The finite-slope support-wise count is $n - a + 1$ by Theorem 7. Every no-loss CA-bad finite slope is support-wise MCA-bad: if the line word is explained on a support S of size at least a and (f, g) were simultaneously explained on that same S, then (f, g) would be C^2-close at the no-loss CA radius. Thus

$$\mathrm{LD}_{\mathrm{ca}}(C, a) \leq n - a + 1.$$

The moving-root line from Theorem 6 gives the matching CA lower bound. Indeed, with $A \subset D$, $|A| = a - 1$, put $B_A(X) = \prod_{x \in A}(X - x)$,

$$f = (XB_A)_{\geq k}, \qquad g = -(B_A)_{\geq k}.$$

For each $t \in D \setminus A$, the slope t is explained on $A \cup \{t\}$. Suppose (f, g) were explained by two degree-$< k$ polynomials on some support T of size at least a. Since

$$|T \cap A| \geq a - (n - a + 1) = 2a - n - 1,$$

and the hypothesis $3a - 2n \geq k$ implies $2a - n - 1 \geq k$ when $a < n$ (the case $a = n$ gives $n - 1 \geq k$), the two explaining polynomials must be the unique degree-$< k$ polynomials agreeing with f and g on A, namely $-(XB_A)_{<k}$ and $(B_A)_{<k}$. At any point $x \in D \setminus A$, simultaneous agreement with those two polynomials would force both $XB_A(x) = 0$ and $B_A(x) = 0$, impossible. Hence (f, g) is globally CA-far at agreement a, and the $n - a + 1$ slopes are CA-bad.

It remains to pass from finite to projective slopes. A projective bad-slope set cannot be all of $\mathbb{P}^1(F)$: otherwise, after choosing any projective point as infinity, the remaining q points would be finite bad slopes in an affine coordinate, contradicting $q > n-a+1$ since $D \subseteq F$ and $a > k$. Choose a nonbad projective point and send it to infinity by a change of basis of the two-dimensional received-line span. All bad projective points are then finite slopes in the new coordinate, so Theorem 7 bounds their number by $n-a+1$. The affine moving-root construction gives the matching projective lower bound. $\square$

20.2 Finite-parameter curve MCA

Fix an integer $d \geq 1$. A degree-d received curve is a family

$$W_\gamma = f_0 + \gamma f_1 + \cdots + \gamma^d f_d, \qquad \gamma \in F.$$

Let $\mathrm{CurveLD}^{(d)}_{\mathrm{sw}}(C, a)$ be the maximum number of parameters $\gamma \in F$ for which W_γ has a C-explanation on a support of size at least a, while the tuple $(f_0, \ldots, f_d)$ is not simultaneously explained by C^{d+1} on that support. Define $\mathrm{CurveLD}^{(d)}_{\mathrm{ca}}(C, a)$ similarly, replacing the support-wise condition by the no-loss global CA condition that $(f_0, \ldots, f_d)$ is not C^{d+1}-close at agreement a.

Theorem 9 (degree-d curve tangent staircase)**.** *Let $C = \mathrm{RS}[F, D, k]$, $|D| = n$, $|F| = q$, and $d \geq 1$. If*

$$(d+2)a - (d+1)n \geq k,$$

then

$$\mathrm{CurveLD}^{(d)}_{\mathrm{sw}}(C, a) = \mathrm{CurveLD}^{(d)}_{\mathrm{ca}}(C, a) = \min\{q,\ d(n-a+1)\}.$$

Proof. Put $m = n-a+1$. For the lower bound, choose $A \subset D$ with $|A| = a-1$ and let $\Omega = D \setminus A$, so $|\Omega| = m$. Choose $M = \min\{q, dm\}$ distinct parameters in F, partition them among the points $x \in \Omega$ into sets R_x of size at most d, and for each $x \in \Omega$ choose a nonzero polynomial

$$P_x(T) = \sum_{i=0}^{d} p_{i,x} T^i$$

whose root set contains R_x. Define $f_i = 0$ on A and $f_i(x) = p_{i,x}$ for $x \in \Omega$. If $\gamma \in R_x$, then W_γ vanishes on $A \cup \{x\}$, so it is explained by the zero codeword on a support of size a. On this same support the tuple $(f_0, \ldots, f_d)$ is not simultaneously explained: every component vanishes on A, which has at least k points, so the only possible explaining codewords are all zero, but P_x is not the zero polynomial and hence some component is nonzero at x.

The same construction is CA-far. If the tuple were globally explained on a support T of size at least a, then

$$|T \cap A| \geq 2a - n - 1.$$

The curve-range hypothesis implies $2a - n - 1 \geq k$ when $a < n$, and the case $a = n$ is immediate from $n-1 \geq k$. Thus all explaining codewords would be zero, which is impossible on every point of Ω by construction. Hence all M parameters are curve-CA-bad and support-wise curve-MCA-bad.

For the upper bound, it is enough to prove the support-wise statement, since curve-CA-bad parameters are support-wise bad. If there are at most d bad parameters, the bound is immediate. Otherwise pick $d+1$ distinct bad parameters $\gamma_0, \ldots, \gamma_d$, with explaining codewords on supports $S_0, \ldots, S_d$. Their intersection has size at least

$$b_0 \geq (d+1)a - dn.$$

On this intersection, interpolation in the parameter γ expresses each component f_i as an F-linear combination of the $d+1$ explaining codewords, hence as a codeword $c_i \in C$. Let S_* be a maximal support on which all $f_i = c_i$, and put $b = |S_*|$. Then $b \geq b_0$, and the hypothesis gives

$$a + b - n \geq a + b_0 - n = (d+2)a - (d+1)n \geq k.$$

Now consider any bad parameter γ, with explaining codeword p_γ on a support S of size at least a. On $S \cap S_*$, whose size is at least $a + b - n \geq k$, the codeword p_γ agrees with

$$c_\gamma = \sum_{i=0}^{d} \gamma^i c_i \in C.$$

Therefore the residual word $W_\gamma - c_\gamma$ must vanish on at least $h = \max\{1, a - b\}$ points of $D \setminus S_*$. By maximality of S_*, for each residual coordinate $x \in D \setminus S_*$ the polynomial

$$R_x(T) = \sum_{i=0}^{d} T^i (f_i(x) - c_i(x))$$

is not identically zero, so it has at most d roots in F. Counting incidences (γ, x) with $R_x(\gamma) = 0$ gives

$$\#\{\text{bad parameters}\} \cdot h \leq d(n - b).$$

If $b \geq a$, then $h = 1$ and the right side is at most $d(n-a) \leq d(n-a+1)$. If $b < a$, then

$$\#\{\text{bad parameters}\} \leq d\,\frac{n-b}{a-b} = d\left(1 + \frac{n-a}{a-b}\right) \leq d(n-a+1),$$

because $a - b \geq 1$. The trivial field-size cap gives the additional upper bound q. This proves the exact formula. $\square$

20.3 Interleaved-list uniqueness

For $\mu \geq 1$, let $\Lambda_\mu(C, a)$ denote the maximum, over all μ-row received words $U = (U_1, \ldots, U_\mu)$, of the number of tuples $(c_1, \ldots, c_\mu) \in C^\mu$ that agree with U on at least a common positions.

Theorem 10 (high-agreement interleaved-list uniqueness). *Let $C \subseteq F^D$ be MDS of length n and dimension k. If*

$$2a - n \geq k,$$

then for every $\mu \geq 1$,

$$\Lambda_\mu(C, a) = 1.$$

Proof. The lower bound is obtained by taking U itself to be a code tuple. For the upper bound, suppose two tuples $c = (c_1, \ldots, c_\mu)$ and $c' = (c'_1, \ldots, c'_\mu)$ both agree with U on supports of size at least a. The supports intersect in at least $2a - n \geq k$ positions. On this intersection, each row c_i agrees with c'_i. Since C is MDS, two codewords that agree on k positions are equal. Thus $c_i = c'_i$ for every row i, so the tuples are identical. $\square$

20.4 The $\mathbb{F}_{17^{32}}$, $n = 512$, $k = 256$ protocol-facing ledger

Corollary 5 (high-agreement adjacent ledgers for the 17^{32} row). *Let $H \subset \mathbb{F}_{17^{32}}^*$ have order 512, and let*

$$C = \mathrm{RS}[\mathbb{F}_{17^{32}}, H, 256].$$

Put $q = 17^{32}$ and

$$B_* = \left\lfloor q/2^{128} \right\rfloor = 6.$$

Then the following statements hold.

1. *For every $a \geq 427$,*

$$\mathrm{LD}_{\mathrm{ca}}(C, a) = \mathrm{LD}_{\mathrm{sw,proj}}(C, a) = \mathrm{LD}_{\mathrm{sw}}(C, a) = 513 - a.$$

   Thus the affine/projective line numerator is 6 at $a = 507$ and 7 at $a = 506$.

2. *For every $\mu \geq 1$ and every $a \geq 384$,*

$$\Lambda_\mu(C, a) = 1.$$

3. *For degree-d finite-parameter curves, whenever*

$$a \geq \left\lceil \frac{(d+1)512 + 256}{d+2} \right\rceil,$$

   we have

$$\mathrm{CurveLD}_{\mathrm{sw}}^{(d)}(C, a) = \mathrm{CurveLD}_{\mathrm{ca}}^{(d)}(C, a) = \min\{17^{32},\ d(513 - a)\}.$$

Proof. The first item is Corollary 3 together with Theorem 8. The second item is Theorem 10, because $2a - 512 \geq 256$ is equivalent to $a \geq 384$. The third item is Theorem 9 after substituting $n = 512$ and $k = 256$. The integer $B_* = 6$ follows from

$$6 \cdot 2^{128} < 17^{32} < 7 \cdot 2^{128}.$$

□

Corollary 6 (conditional high-agreement protocol ledger). *Consider a protocol reduction for the row of Corollary 5 whose coding-theoretic error at agreement a is exactly*

$$\frac{N_{\mathrm{geom}}(a)}{17^{32}} + \frac{\Lambda_\mu(C, a)}{17^{32}} + \varepsilon_{\mathrm{query}},$$

where $N_{\mathrm{geom}}(a)$ is one affine/projective line CA/MCA numerator or one degree-d finite-parameter curve numerator, and where no additional fold, hash, Fiat–Shamir, or extension-lift loss is hidden in the notation. In the high-agreement ranges above, this ledger becomes the following explicit integer test.

For affine/projective line CA/MCA,

$$N_{\mathrm{geom}}(a) + \Lambda_\mu(C, a) = (513 - a) + 1 = 514 - a \qquad (a \geq 427).$$

With $\varepsilon_{\mathrm{query}} = 0$, this is at most the 2^{-128} budget iff $a \geq 508$. Thus the combined line-plus-list ledger is safe at agreement 508 and unsafe at agreement 507.

For degree-d finite-parameter curve MCA/CA,

$$N_{\mathrm{geom}}(a) + \Lambda_\mu(C,a) = d(513-a)+1$$

in the curve-staircase range. With $\varepsilon_{\mathrm{query}} = 0$*, the safe grid points are exactly those satisfying*

$$d(513-a)+1 \le 6.$$

In particular:

d	*curve-plus-list safe grid*	*curve term alone safe grid*
1	$a \ge 508$ $(r \le 4)$	$a \ge 507$ $(r \le 5)$
2	$a \ge 511$ $(r \le 1)$	$a \ge 510$ $(r \le 2)$
3	$a = 512$ $(r = 0)$	$a \ge 511$ $(r \le 1)$
4	$a = 512$ $(r = 0)$	$a = 512$ $(r = 0)$
5	$a = 512$ $(r = 0)$	$a = 512$ $(r = 0)$
6	*no safe grid point with the list term*	$a = 512$ $(r = 0)$
$d \ge 7$	*no safe grid point with the list term*	*no safe grid point for the curve term alone*

where $r = n - a$ *is the integer Hamming radius.*

Proof. By Corollary 5, the interleaved-list numerator is 1 throughout the displayed high-agreement range. The affine/projective line numerator is $513-a$, and the degree-d curve numerator is $d(513-a)$. Since

$$\left\lfloor 17^{32}/2^{128} \right\rfloor = 6,$$

the code-level numerator must be at most 6 when $\varepsilon_{\mathrm{query}} = 0$. The displayed inequalities are exactly the resulting integer conditions. □

Remark 7 (what this does and does not settle)**.** *Theorems 8, 9, and 10 are unconditional finite statements under the hypotheses printed above: the CA/projective/curve statements are Reed–Solomon evaluation-domain statements, while interleaved uniqueness is MDS. Corollary 6 is conditional on the protocol really consuming only the printed coding terms over the printed field. It does not cover reductions with additional curve families, extension-line lifts, folding losses, query errors, or cryptographic errors unless those terms are added to the ledger.*

sectionTowards-Prize Finite-Threshold Theorems

Status: New-local finite theorem infrastructure plus Conditional strict264 gate. The results in this section are deliberately finite and certificate-facing. They are meant to move the experimental ledger from lower-bound numerators toward the integer agreement staircase needed by the Proximity Prize. They do not prove the corrected Paper B local-limit conjecture. The later high-agreement tangent staircase supersedes the old strict264/265 search target as a threshold-pinning route for the $\mathbb{F}_{17^{32}}$, $n = 512$, $k = 256$ row; the strict264 material below remains useful as certificate language and as a record of quotient-core mechanisms.

For a received line $\ell_z = f + zg$, write

$$\mathrm{Bad}_{f,g}(C,a)$$

for the set of finite slopes $z \in F$ for which there is a support $S \subseteq D$, $|S| \ge a$, and a codeword $c \in C$ satisfying $c|_S = (f+zg)|_S$, but there is no pair of codewords $c_f, c_g \in C$ with $c_f|_S = f|_S$ and $c_g|_S = g|_S$. Put

$$\mathrm{LD}^{f,g}_{\mathrm{sw}}(C,a) = |\,\mathrm{Bad}_{f,g}(C,a)|, \qquad \mathrm{LD}_{\mathrm{sw}}(C,a) = \max_{f,g} \mathrm{LD}^{f,g}_{\mathrm{sw}}(C,a),$$

where the maximum is over the line family allowed by the sampler under consideration. The field paying the denominator is always q_{line}, not silently q_{gen} or q_{chal}.

Proposition 8 (certificate-to-$\mathrm{LD}_{\mathrm{sw}}$ gate)**.** *Let $C = \mathrm{RS}[F, D, k]$, let $a \geq k$, and fix a received line $f + zg$. Suppose there is a finite set $\mathcal{C}$ of tuples*

$$(z_i, S_i, P_i), \qquad z_i \in F, \quad S_i \subseteq D, \quad |S_i| \geq a, \quad P_i \in F[X]_{<k},$$

with the following properties:

1. *the slopes z_i are pairwise distinct;*
2. *$P_i(x) = f(x) + z_i g(x)$ for every $x \in S_i$;*
3. *there are no $P_{i,f}, P_{i,g} \in F[X]_{<k}$ satisfying $P_{i,f}|_{S_i} = f|_{S_i}$ and $P_{i,g}|_{S_i} = g|_{S_i}$.*

Then

$$\mathrm{LD}^{f,g}_{\mathrm{sw}}(C, a) \geq |\mathcal{C}|, \qquad \mathrm{LD}_{\mathrm{sw}}(C, a) \geq |\mathcal{C}|.$$

Proof. Each tuple certifies that the slope z_i is explained on a support of size at least a, and the third condition is exactly support-wise noncontainment on that support. Pairwise distinctness of the slopes prevents multiple tuples from paying for the same bad slope. Thus all z_i's lie in $\mathrm{Bad}_{f,g}(C, a)$, giving the first inequality. The second follows from the definition of $\mathrm{LD}_{\mathrm{sw}}(C, a)$ as a maximum over lines. □

Corollary 7 (strict264 certificate target)**.** *For the row*

$$C = \mathrm{RS}[\mathbb{F}_{17^{32}}, H, 256], \qquad |H| = 512,$$

a strict264 certificate consists of seven tuples in Proposition 8 with $a = 264$. Such a certificate proves

$$\mathrm{LD}_{\mathrm{sw}}(C, 264) \geq 7.$$

Proof. Apply Proposition 8 with $|\mathcal{C}| = 7$ and $a = 264$. □

Proposition 9 (fixed-locator unique-slope principle)**.** *Use the notation of Theorem 4. Fix $a \geq k$, put $t = a - k$, $j = n - a = r - t$, and fix a squarefree D-split locator L_T of degree j. Let*

$$A_T = H_{t,j}(u)\boldsymbol{\ell}_T, \qquad B_T = H_{t,j}(v)\boldsymbol{\ell}_T.$$

If $B_T \neq 0$, then this locator contributes at most one finite support-wise noncontained slope. More precisely, a slope exists exactly when A_T is a scalar multiple of B_T, in which case the unique slope is

$$z_T = -A_{T,m}/B_{T,m}$$

for any coordinate m with $B_{T,m} \neq 0$, and the same value must be obtained from every nonzero coordinate of B_T. Consequently

$$\mathrm{LD}^{f,g}_{\mathrm{sw}}(C, a) \leq \#\{T \subset D : |T| = n - a, L_T \text{ squarefree and } D\text{-split}, B_T \neq 0, A_T \in F B_T\}.$$

Proof. For the fixed locator, Theorem 4 gives the incidence equation

$$A_T + zB_T = 0$$

and the noncontainment condition $B_T \neq 0$. If $B_T \neq 0$, any solution z is forced by any coordinate where B_T is nonzero. Existence is exactly the consistency condition that all coordinates give the same scalar, equivalently $A_T \in FB_T$. Counting possible locators gives the final bound; different locators may still collide to the same slope, so the displayed quantity is an upper bound on distinct slopes. □

Remark 8 (why this helps the 265 upper bound)**.** *For the current $n = 512, k = 256$ row, this proposition remains useful for duplicate-aware locator searches. The old proposed upper bound $\mathrm{LD}_{\mathrm{sw}}(C, 265) \leq 6$ is now known to be false by Theorem 6, which already gives $\mathrm{LD}_{\mathrm{sw}}(C, 265) \geq 248$. Future upper-bound searches should therefore start beyond the exact tangent gate, at agreement 507 and higher, or explicitly isolate non-tangent mechanisms.*

Theorem 11 (subfield confinement for base-valued lines)**.** *Let $K \subseteq F$ be a subfield, let $D \subset K$, and let $C_F = \mathrm{RS}[F, D, k]$. Suppose $f, g : D \to K$, and let $S \subseteq D$ have $|S| \geq k$. If $z \in F \setminus K$ and there is a polynomial $P \in F[X]_{<k}$ with*

$$P(x) = f(x) + zg(x) \qquad (x \in S),$$

then there are polynomials $P_f, P_g \in K[X]_{<k}$ such that

$$P_f|_S = f|_S, \qquad P_g|_S = g|_S.$$

Consequently, for K-valued received lines and agreement $a \geq k$, every support-wise noncontained finite slope lies in K.

Proof. Choose a subset $S_0 \subseteq S$ of size k. Interpolate the values of f and g on S_0 to obtain unique polynomials $P_f, P_g \in K[X]_{<k}$. On S_0, the polynomial $P_f + zP_g$ agrees with P. Since both have degree less than k and agree on k points, they are equal as polynomials over F. For every $x \in S$, therefore,

$$0 = f(x) + zg(x) - P(x) = (f(x) - P_f(x)) + z(g(x) - P_g(x)).$$

Both coefficients in the final expression lie in K. Since $z \notin K$, the elements $1, z$ are linearly independent over K. Hence both coefficients vanish for every $x \in S$, proving the separate explanations of f and g. Thus any explained slope outside K is support-wise contained, so a noncontained slope must lie in K. □

Remark 9 (extension-field warning)**.** *Theorem 11 is intentionally one-sided. It only treats K-valued received lines. It does not rule out genuinely F-valued residue-line witnesses, and therefore must not be used as an extension-field MCA theorem without an additional hypothesis forcing the line to be K-valued.*

Theorem 12 (seven-slope arithmetic gate over $\mathbb{F}_{17^{32}}$)**.** *Let $q_{\mathrm{line}} = 17^{32}$. If a row over $\mathbb{F}_{17^{32}}$ has $\mathrm{LD}_{\mathrm{sw}}(C, 264) \geq 7$, then at radius*

$$\delta = 1 - \frac{264}{512} = \frac{31}{64}$$

its support-wise MCA error is strictly larger than 2^{-128}. In particular, for the $n = 512, k = 256$ row, any strict264 seven-slope certificate proves

$$\varepsilon_{\mathrm{mca}}(C, 31/64) > 2^{-128}.$$

Proof. The MCA sampler over finite slopes has denominator $q_{\mathrm{line}} = 17^{32}$. Seven bad slopes therefore give error at least $7/17^{32}$. The exact integer comparison is

$$7 \cdot 2^{128} - 17^{32} = 140649736861019983995159547014262328 31 > 0.$$

Thus $7/17^{32} > 2^{-128}$. Agreement 264 on a domain of size 512 corresponds to relative radius $1 - 264/512 = 31/64$, giving the claim. $\square$

Theorem 13 (one-step staircase pinning criterion)**.** *Fix a finite-slope sampler of denominator q_{line} and a target $\varepsilon_* = 2^{-128}$. Let*

$$B(a) = \mathrm{LD}_{\mathrm{sw}}(C, a)$$

for the allowed line family, and let $\delta_a = 1 - a/n$. If, for some integer $a \geq k$,

$$B(a) > \varepsilon_* q_{\mathrm{line}} \qquad \textit{and} \qquad B(a+1) \leq \varepsilon_* q_{\mathrm{line}},$$

then the closed-ball agreement staircase is unsafe at radius δ_a and safe on the next open interval below it, including the grid radius δ_{a+1}. Equivalently, the largest grid radius certified safe is δ_{a+1}, while the continuous supremum of certified safe radii is δ_a and is not attained under the closed endpoint convention.

Before the tangent staircase was isolated, the special case

$$B(264) \geq 7, \qquad B(265) \leq 6$$

would have pinned the finite row to an unsafe/safe transition between

$$\delta_{265} = 247/512 \qquad \textit{and} \qquad \delta_{264} = 31/64.$$

For the $\mathbb{F}_{17^{32}}$, $n = 512$, $k = 256$ row, the premise $B(265) \leq 6$ is false by the moving-root tangent floor. The actual finite-slope staircase gate supplied by Corollary 3 is $B(506) = 7$, $B(507) = 6$.

Proof. The function $B(a)$ is nonincreasing in a, because raising the agreement threshold only removes valid supports. At radius δ_a, the closed-ball predicate allows agreement at least a, so the error is $B(a)/q_{\mathrm{line}}$, which is larger than ε_* by assumption. At radius δ_{a+1}, the predicate requires agreement at least $a + 1$, so the error is $B(a + 1)/q_{\mathrm{line}}$, which is at most ε_*.

More generally, any radius $\delta < \delta_a$ sufficiently close to δ_a has integer distance bound at most $n - a - 1$, hence agreement at least $a + 1$. Thus the closed endpoint at δ_a is the first unsafe grid step, while the safe side is immediately below that endpoint. For $q_{\mathrm{line}} = 17^{32}$, the inequalities

$$6 \cdot 2^{128} < 17^{32} < 7 \cdot 2^{128}$$

convert the printed integer conditions into the two displayed error inequalities, and $1 - 265/512 = 247/512$, $1 - 264/512 = 31/64$. $\square$

21 Non-Claims

This note does not claim:

- a proof of the corrected MCA conjecture;
- a positive theorem above corrected reserve;

- a q_{gen} local-limit theorem;
- a list, CA, MCA, line-decoding, SNARK, or protocol consequence;
- a large $\Theta(p^2)$ counterpacket;
- a new prize-worthy numerator beyond the current Cycle116/119 52,747,567,092 finite/source-scoped record;
- priority for the BCHKS quotient-locator proof pattern.

The mathematical value is narrower. On the imported side, the file gives a provenance-safe dictionary and wrapper for using BCHKS without claiming it as new. On the local Paper B side, the possible large branch in the restricted quadratic-extension window must satisfy an explicit algebraic identity before it can even be surface-sized.

A BibTeX Entry to Copy into Main Papers

```
@misc{BCHKS25,
  author = {Eli Ben-Sasson and Dan Carmon and Ulrich Habock and
            Swastik Kopparty and Shubhangi Saraf},
  title  = {On Proximity Gaps for Reed--Solomon Codes},
  year   = {2025},
  note   = {Manuscript dated November 11, 2025},
  url    = {https://www.math.toronto.edu/swastik/rs-proximity-gaps-2025.pdf}
}
```

References

[1] The Proximity Prize. *The Proximity Prize: Reed–Solomon Challenge*. Preliminary version, accessed June 26, 2026. `https://proximityprize.org/`.

[2] Eli Ben-Sasson, Dan Carmon, Ulrich Haböck, Swastik Kopparty, and Shubhangi Saraf. *On Proximity Gaps for Reed–Solomon Codes*. Manuscript dated November 11, 2025. `https://www.math.toronto.edu/swastik/rs-proximity-gaps-2025.pdf`.